

Lipids and Cerebrovascular Disease

Objectives

- Review lipoprotein biochemistry
- Lipids and cerebrovascular disease
- Lipid management in stroke
- Emerging lipid biomarkers
- Future directions

Lipids

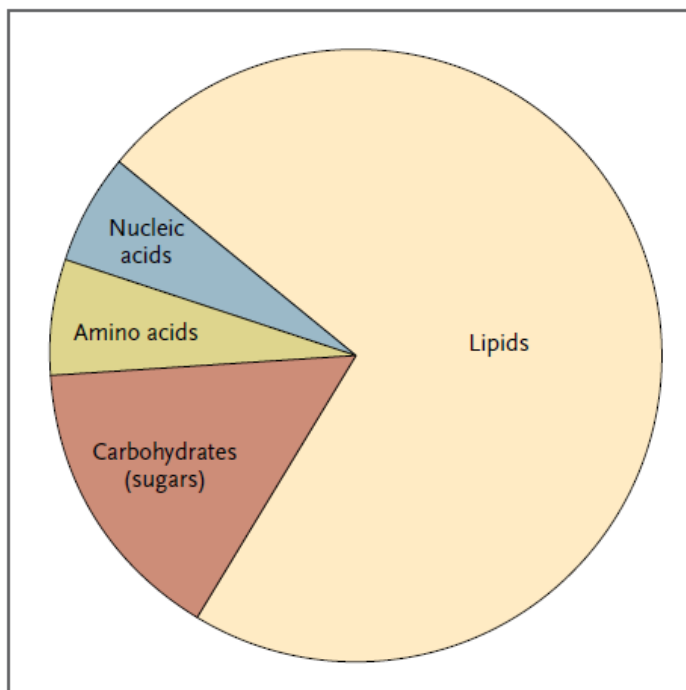


Figure 1. Relative Distribution of Biologic Molecules in Human Plasma.

Amino acids and nucleic acids are shown without consideration of the contribution of proteins and DNA or RNA. The relative distribution is based on weight (grams per deciliter). Data were compiled from Lentner,¹ Wishart et al.,² and Quehenberger et al.³

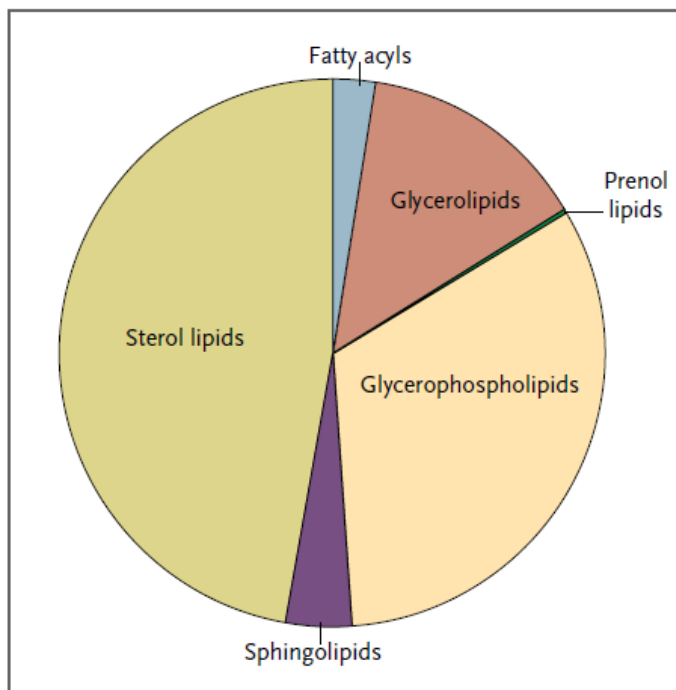
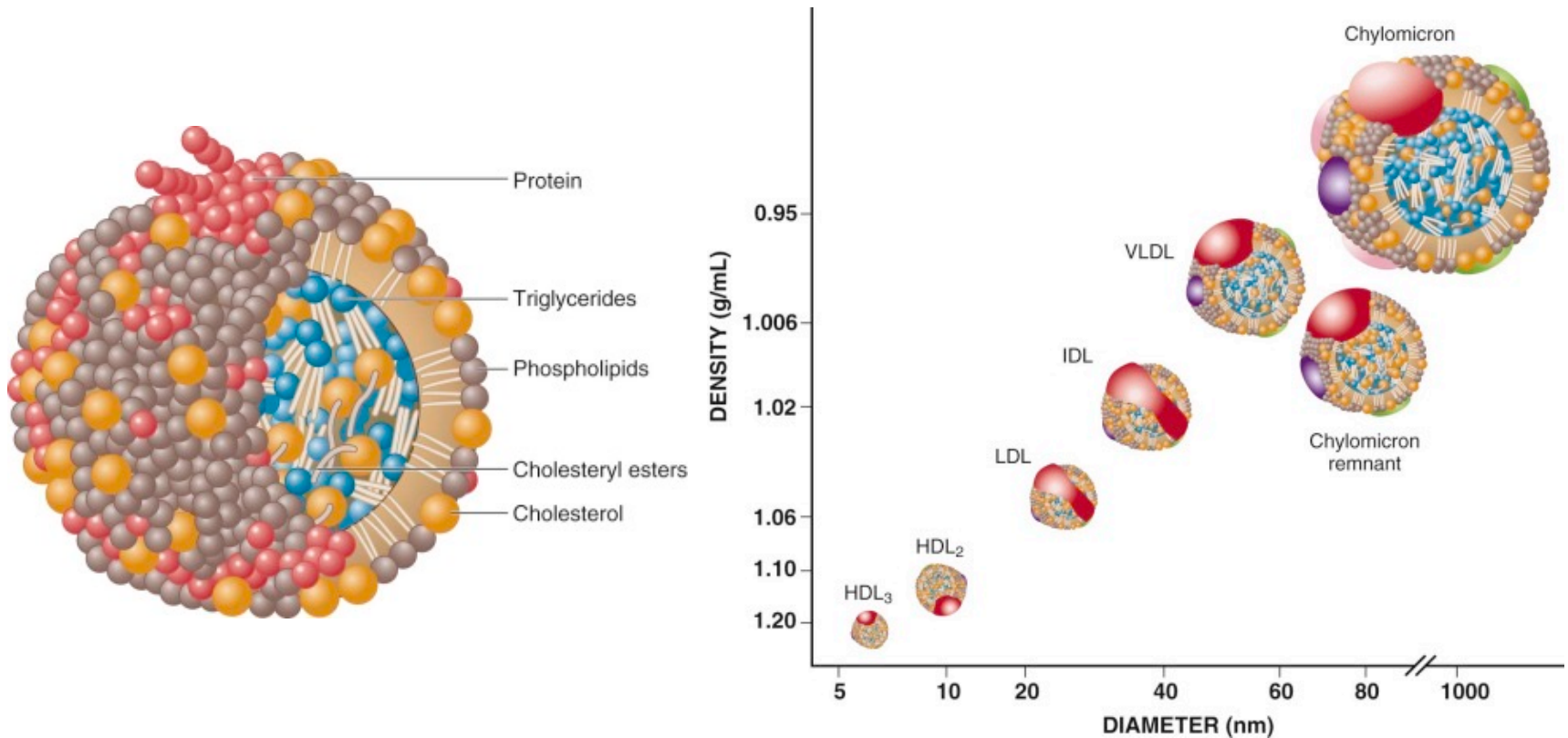


Figure 2. Relative Distribution of Lipids in Human Plasma.

Lipidomic analysis has identified, characterized, and quantified almost 600 lipid molecular species in human plasma.³ The relative distribution in each category is given on a molar basis. The nomenclature of the lipid categories conforms to the recently developed LIPID MAPS classification system.⁶

Lipoproteins



Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. Genest, Jacques; Libby, Peter.

Source-qualified teaching update | 30 September 2026

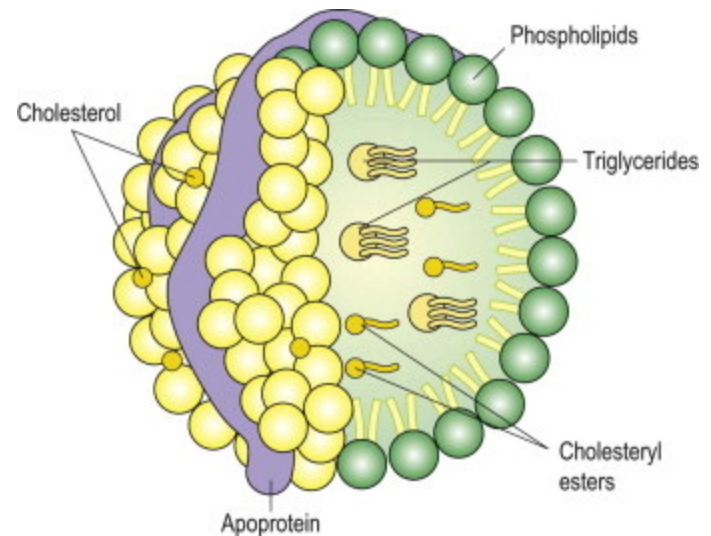
The Friedewald TG/5 equation requires mg/dL concentrations (use TG/2.2 with mmol/L). It should not be used with TG $\geq$ 400 mg/dL; the 2026 guideline prefers Martin/Hopkins or Sampson/NIH estimation where appropriate.

doi.org/10.1161/CIR.0000000000001423

pubmed.ncbi.nlm.nih.gov/4337382/

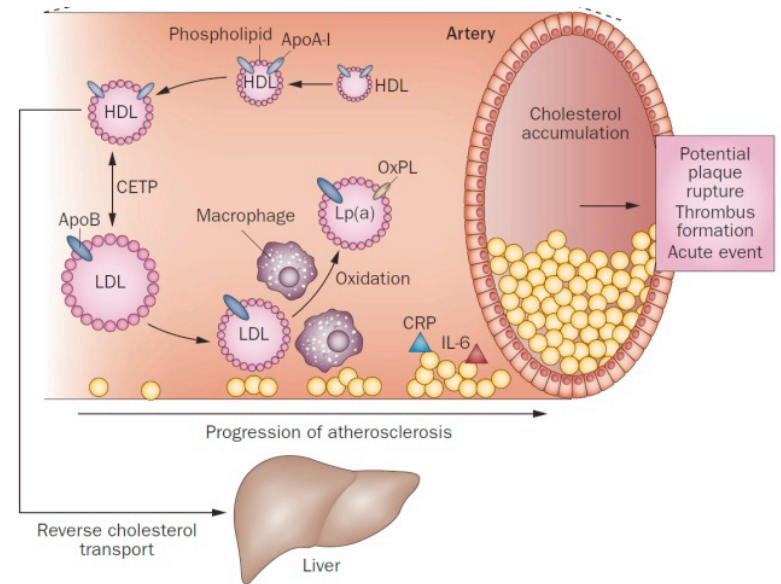
Apoproteins

- Functions
 - Lipoprotein assembly & secretion
 - Lipoprotein structural integrity
 - Enzyme co-activation or inhibition
 - Binding/docking to specific receptors/proteins



Particle	Density (kg/L)	Main component	Apolipoproteins *	Diameter (nm)
Chylomicrons	0.95	TG	B48 (A, C, E)	75–1200
VLDL	0.95–1.006	TG	B100 (A, C, E)	30–80
IDL	1.006–1.019	TG & cholesterol	B100, E	25–35
LDL	1.019–1.063	Cholesterol	B100	18–25
HDL	1.063–1.210	Protein	AI, AII (C, E)	5–12

- Apo A1
 - ~70% of HDL protein
 - Role: structural, lecithin-cholesterol acyltransferase activator
 - Marker of HDL concentration
- Apo B100
 - LDL, VLDL/IDL
 - Role: structural, ligand for LDL receptor



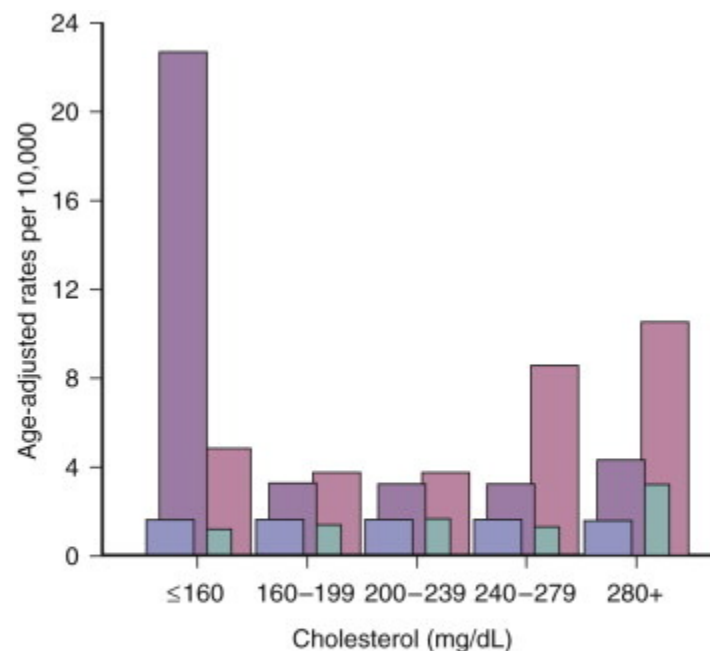
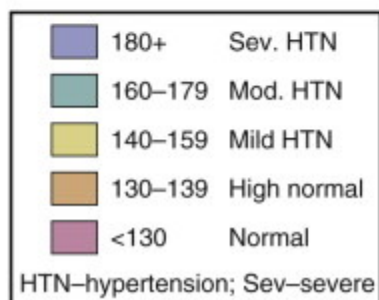
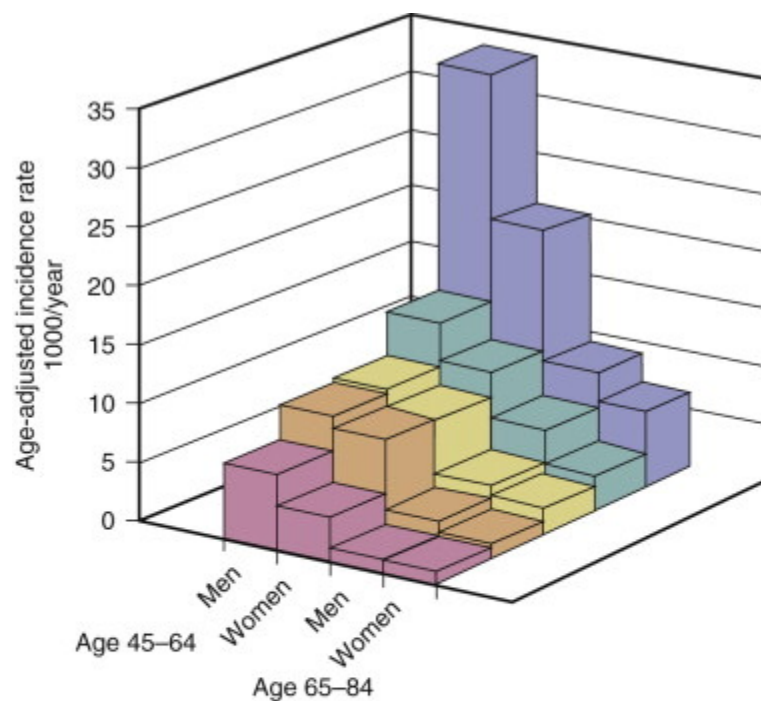
Medical Biochemistry. Dominiczak, Marek H.
 Nat Rev Endocrinol. 2014 Nov;10(11):659-72.

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ApoB100 occurs on VLDL, IDL and LDL. Chylomicron remnants contain ApoB48; distinguish these isoforms from a total ApoB measurement.

pubmed.ncbi.nlm.nih.gov/25178732/
pubmed.ncbi.nlm.nih.gov/9215545/

Epidemiology



Total Cholesterol and LDL-C

Lipid Profile Component	Study	Results
Total cholesterol	Alpha-Tocopherol, Beta Carotene Cancer Prevention Study ¹	Increased risk of ischemic stroke with a total cholesterol levels ≥ 7 mmol/L (≥ 271 mg/dL; HR, 1.25; 95% CI, 0.99–1.57)
	Asia Pacific Cohort Studies Collaboration ²	For every 1-mmol/L increase in total cholesterol, there was a 25% (95% CI, 13–40) increase in ischemic stroke rate
	Women's Health Study ³	Total cholesterol level was associated with ischemic stroke (adjusted HR per 1-mmol/L increase for ischemic stroke of 1.17 (95% CI, 1.06–1.30)
	Eurostroke Project ⁴	No relationship between total cholesterol level and ischemic stroke
LDL cholesterol	Women's Pooling Project ⁶	Higher total cholesterol was associated with increased risk of ischemic stroke (HR, 1.69; 95% CI, 0.91–3.13, for topmost quintile compared with the lower-most one)
	Women's Health Study ³	Serum LDL cholesterol was associated with increased risk of ischemic stroke (HR, 1.74; 95% CI, 1.14–2.66; P (trend across quintiles)=0.003)
	The Atherosclerosis Risk in Communities ⁵	Nonsignificant association between LDL cholesterol and ischemic stroke (HR, 1.26; 95% CI, 0.91–1.76, for topmost quintile compared with the lower-most one)

	Age-adjusted HR (95% CI)
Total Cholesterol (mg/dL)[§]	
<179	1.00 (referent)
179-199	1.69 (1.03-2.77)
200-218	1.78 (1.09-2.88)
219-243	2.20 (1.38-3.49)
≥ 244	2.40 (1.52-3.80)
P for trend	<0.001
LDL Cholesterol (mg/dL)[§]	
<96	1.00 (referent)
96-113	1.36 (0.86-2.14)
114-129	1.19 (0.75-1.89)
130-150	1.97 (1.30-2.98)
≥ 151	1.85 (1.22-2.80)
P for trend	<0.001

↑ TChol & LDL-C α ↑ Ischemic Stroke

Stroke. 2015 Nov;46(11):3322-8.

Neurology. 2007 February 20;68(8):556–562.

Lipid Profile Component	Study	Results
Total cholesterol	Multiple Risk Factor Intervention Trial ¹³	Three-fold increase in the risk of fatal intracerebral hemorrhage in patients total serum cholesterol <4.13 mmol/L when compared with those with values ≥4.13 mmol/L
	Asia Pacific Cohort Studies Collaboration ²	For every 1-mmol/L (38.7 mg/dL) increase in total cholesterol, there was a 20% (95% CI, 8–30) reduction in the risk of hemorrhagic stroke
	Korean Medical Insurance Corporation Study ¹⁴	Low serum cholesterol was not associated with intracerebral hemorrhage
LDL Cholesterol	Pooled cohort analysis of the ARIC study and the Cardiovascular Health Study ¹⁵	LDL was inversely associated with hemorrhagic stroke risk (HR for topmost quartile versus lowest 3 quartiles 0.52; 95% CI, 0.31–0.88)

↓ Total Cholesterol & LDL-C α
↑ Hemorrhagic Stroke

Table 3. Multivariable Adjusted Independent Associations of Baseline Risk Factors With Intraparenchymal Hemorrhage, Combined ARIC/CHS Cohort

Variable (contrast)	Relative Rate	95% CI	P Value
Age per 10 years	2.00	1.69, 2.38	<0.0001
Hypertension Status*			
Normal or High Normal	1.00		
Stage 1 Hypertension	1.43	0.90, 2.26	0.132
Stage 2 Hypertension	2.71	1.58, 4.67	0.0003
Stage 3 Hypertension	5.55	3.07, 10.03	<0.0001
Ethnicity (African-American vs Whites)	1.89	1.28, 2.80	0.0019
Triglycerides per log unit, mg/dl†	0.56	0.37, 0.84	0.0039
LDL-C (top 1/4 vs lower 3/4)	0.52	0.31, 0.88	0.0085

Stroke. 2015 Nov;46(11):3322-8.
Stroke. 2013 Jul;44(7):1833-9.
Stroke. 2007;38:2718-2725.

HDL-C

Copenhagen City Heart Study⁷ 47% reduction in the risk of nonhemorrhagic stroke for every 1-mmol/L increase in HDL

Cardiovascular Health Study⁹ Higher HDL cholesterol level was associated with a decreased risk of ischemic stroke in men, but not in women

The Atherosclerosis Risk in Communities⁵ No relationship between HDL cholesterol and ischemic stroke in men and a nonsignificant association in women

Estimates RR per 10 mg/dl increase in HDL-C

Cohort studies

- Population-based sample
 - Walldius et al, 2006
 - Simons et al, 2006
 - Psaty et al, 2004
 - Lindstrom et al, 1994
 - Tanne et al, 1997
- Record-based sample
 - Koren-Morag et al, 2002

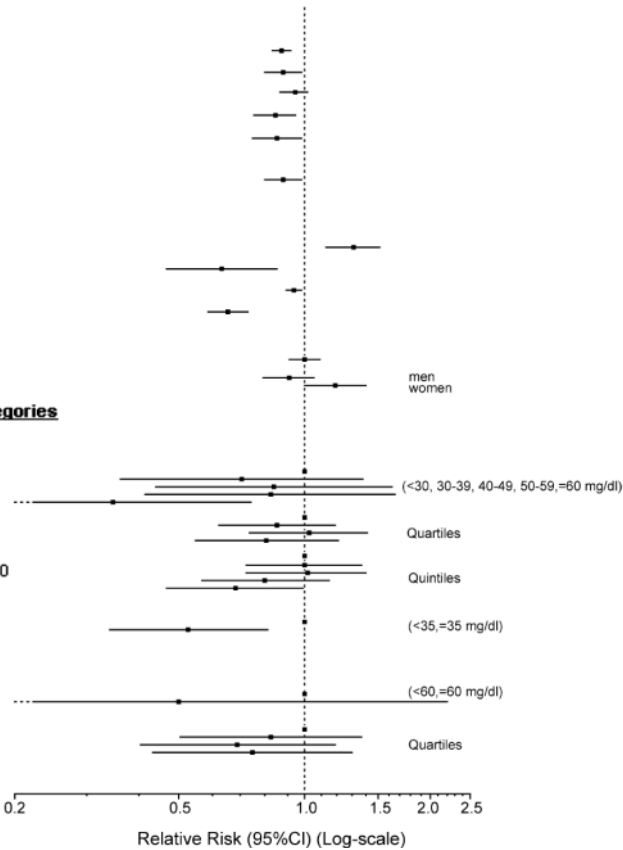
Case-Control studies

- Non-nested design
 - Amarencu et al, 2006
 - Milionis et al, 2005
 - Tirschwell et al, 2004
 - Sacco et al, 2001
- Nested design
 - Patel et al, 2005
 - Bots et al, 2002

Estimates RR per HDL-C categories

Cohort studies

- Population-based sample
 - Soyama et al, 2003
 - Shahar et al, 2003
 - Wannamethee et al, 2000
- Record-based sample
 - Lehto et al, 1996
- Case-Control studies
 - Non-nested design
 - Woo et al, 1991
 - Nested design
 - Bowman et al, 2003



↑ HDL-C α ↓ Ischemic Stroke

Stroke. 2015 Nov;46(11):3322-8.
Atherosclerosis. 2008 Feb;196(2):489-96.

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Northern Manhattan Study: HDL-C ≥ 35 mg/dL, not ≤ 35 , was associated with adjusted OR 0.53 (95% CI 0.39-0.72) for ischemic stroke (539 cases, 905 controls). The reproduced secondary table inherited the reversed threshold; that row has been removed.

pubmed.ncbi.nlm.nih.gov/11386928/
pubmed.ncbi.nlm.nih.gov/26451029/

Triglycerides

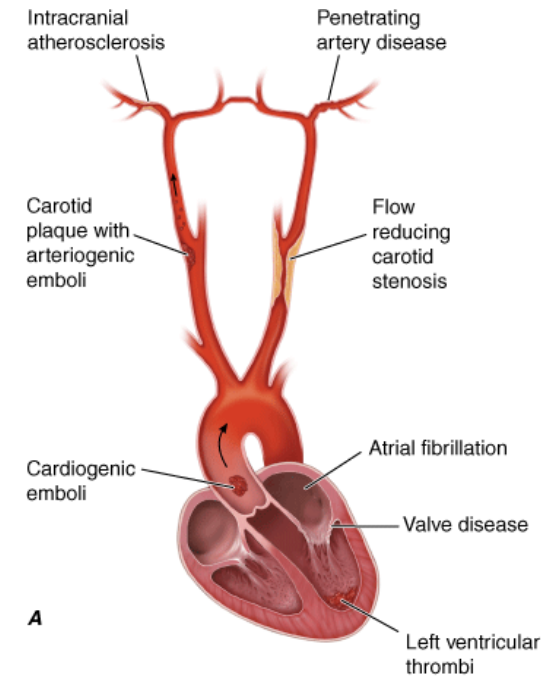
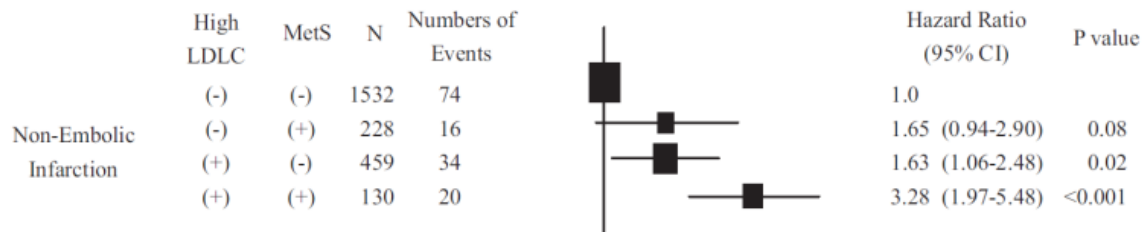
The Atherosclerosis Risk in Communities ⁵	Fasting triglyceride levels were not associated with ischemic stroke
Physicians' Health Study ¹⁰	No association between triglyceride level and ischemic stroke
Copenhagen City Heart Study ¹¹	15% (95% CI, 9–22) increased risk of ischemic stroke for each 89-mg/dL increase in nonfasting triglycerides
Women's Health Study ¹²	The highest tertile of nonfasting but not fasting triglyceride level was associated with increased ischemic stroke risk when compared with lowest tertile
Three City Study ¹⁶	A triglycerides level ≤ 0.94 mmol/L was associated with an increased risk of hemorrhagic stroke (adjusted HR, 2.35; 95% CI, 1.18–4.70)
Rotterdam study ¹⁷	The highest quartile of triglyceride level had a lower risk of intracerebral hemorrhage than the lowest quartile (HR, 0.20; 95% CI, 0.06–0.69)

Triglycerides α $\uparrow$ Ischemic Stroke
 $\downarrow$ Triglycerides α $\uparrow$ Intracerebral Hemorrhage

Lipids and Ischemic Stroke Subtypes

Table 3 OR (95% CIs) for ischemic stroke by quintile of total and HDL cholesterol

Model	Quintile				
	1	2	3	4	5
All ischemic stroke					
Total cholesterol					
Adjusted for baseline covariates*	1.0 (ref)	0.9 (0.7–1.1)	1.0 (0.8–1.2)	0.9 (0.7–1.1)	1.4 (1.2–1.7)
Fully adjusted†	1.0 (ref)	1.0 (0.8–1.2)	1.1 (0.9–1.4)	1.0 (0.8–1.2)	1.6 (1.3–1.9)
Ischemic stroke subtypes					
Total cholesterol‡					
Lacunar/small artery stroke	1.0 (ref)	1.3 (0.8–2.1)	1.3 (0.8–2.2)	1.1 (0.7–1.8)	2.2 (1.4–3.4)
N cases	34	40	40	33	63
Atherosclerotic/large artery stroke	1.0 (ref)	1.0 (0.5–2.0)	1.8 (1.0–3.4)	1.7 (0.9–3.1)	3.1 (1.7–5.6)
N cases	20	16	28	26	41
Embolic stroke	1.0 (ref)	0.9 (0.6–1.4)	1.1 (0.7–1.6)	0.9 (0.6–1.4)	1.2 (0.8–1.8)
N cases	63	56	59	51	56
Other stroke	1.0 (ref)	0.9 (0.7–1.2)	1.0 (0.7–1.3)	0.9 (0.7–1.2)	1.4 (1.0–1.8)
N cases	117	105	112	113	169



Source: Fauci AS, Kasper DL, Braunwald E, Hauser SL, Longo DL, Ja Harrison's Principles of Internal Medicine, 17th Edition: <http://www.acco>

Lipids and Carotid Atherosclerosis

Table 3. OR (95% CIs) for Lipid Core Presence With Cardiovascular Risk Factors (N=214)

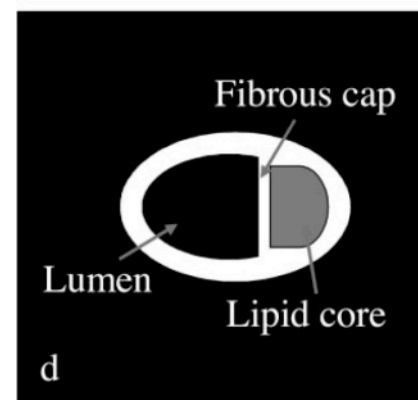
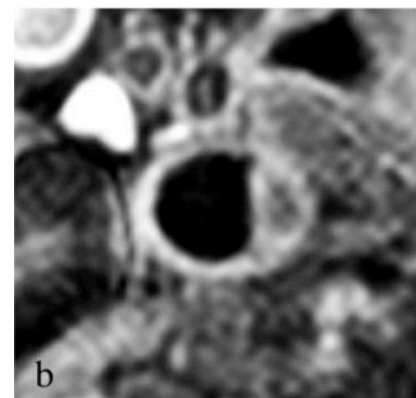
	Adjusted* OR	95% CI
Women vs men	1.35	0.56–3.25
Black vs white	0.52	0.16–1.76
Hispanic vs white	0.77	0.27–2.24
Chinese vs white	2.07	0.49–8.78
Age 45–54 vs >74	0.25	0.06–1.05
55–64 vs >74	0.93	0.28–3.12
65–74 vs >74	0.65	0.20–2.08
Maximum wall thickness per 100 μ m	1.28	1.18–1.40
Total cholesterol	2.76†	1.01–7.51
181–210 mg/dL vs $\leq$ 180 mg/dL		
>210 mg/dL vs $\leq$ 180 mg/dL	4.63‡	1.56–13.75
Smoking: current vs never smoked	0.63	0.16–2.49
Former vs never smoked	0.83	0.33–2.10
Diabetes or IFG vs no diabetes or IFG	0.79	0.28–2.18
Hypertension vs nonhypertensives	1.07	0.43–2.67
CRP: 1.14–2.92 mg/L vs <1.14 mg/L	0.53	0.18–1.53
>2.92 mg/L vs <1.14 mg/L	0.97	0.33–2.90

*Adjusted for all listed variables.

†Reduced to 2.43 (95% CI: 0.87 to 6.75) after adjusting for lipid-lowering medication.

‡Reduced to 4.23 (95% CI: 1.40 to 12.78) after adjusting for lipid-lowering medication.

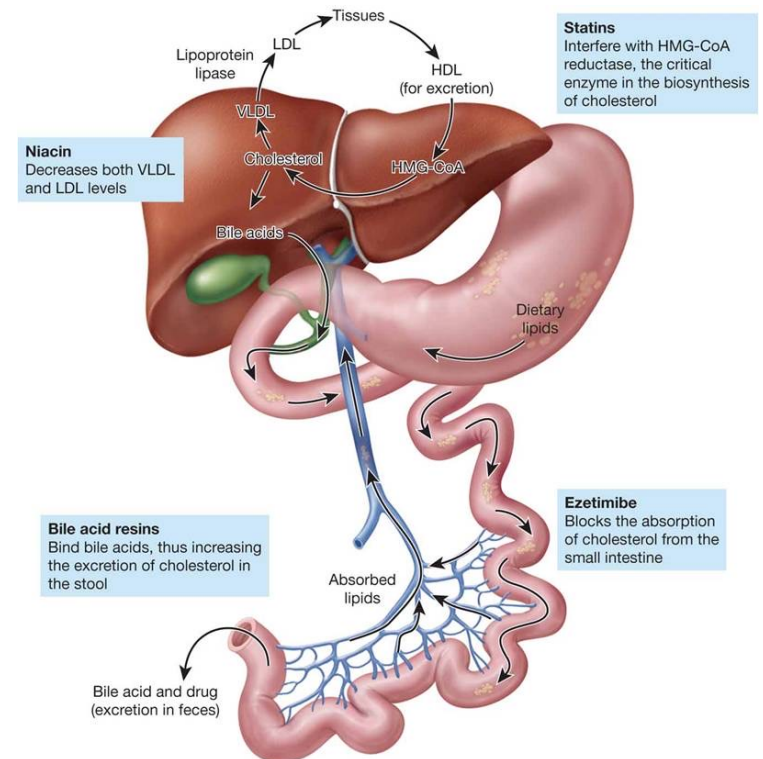
IFG indicates impaired fasting glucose.



Lipid-Lowering Therapies and Stroke

Historical lipid-percentage table retired.

Use indication-specific outcome evidence.
See the source context below.



www.medscape.com

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The unverified historical lipid-percentage treatment table has been retired. Biochemical lipid changes are not equivalent to evidence of stroke-outcome benefit; choose therapy using current indication-specific outcome evidence.

doi.org/10.1161/CIR.0000000000001423

Treatment of Dyslipidemia: Primary Stroke Prevention

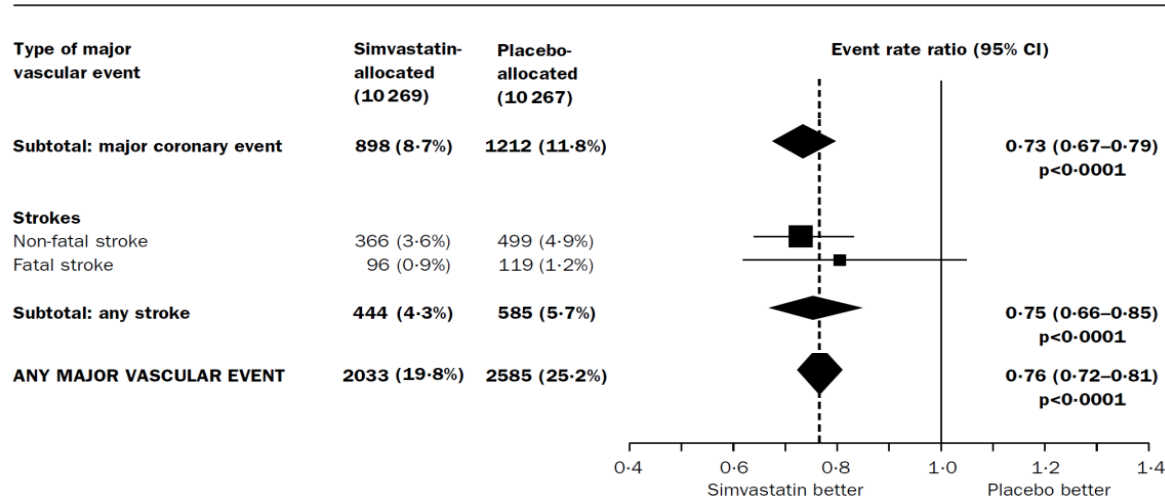
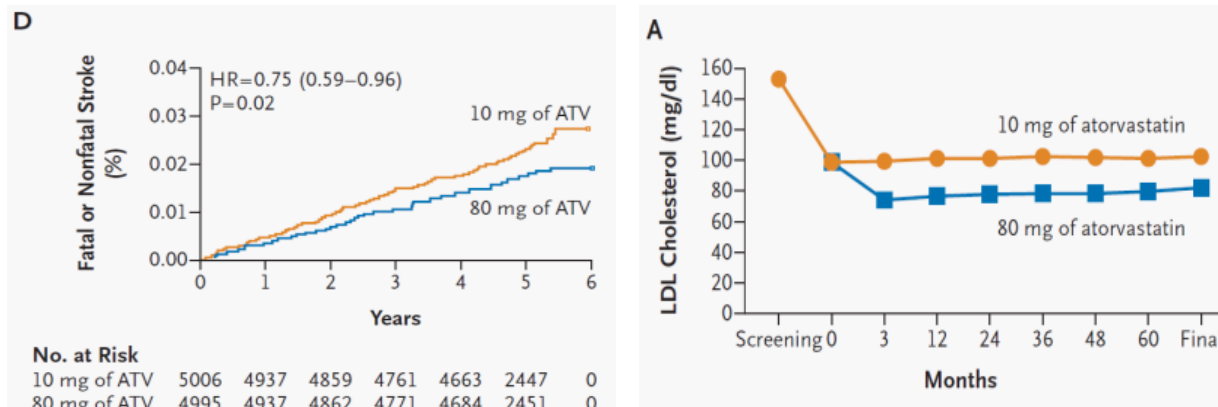


Figure 3: Effects of simvastatin allocation on first major coronary event, stroke, and revascularisation (defined prospectively as “major vascular events”)



Lancet. 2002 Jul 6;360(9326):7-22.

N Engl J Med. 2005 Apr 7;352(14):1425-35.

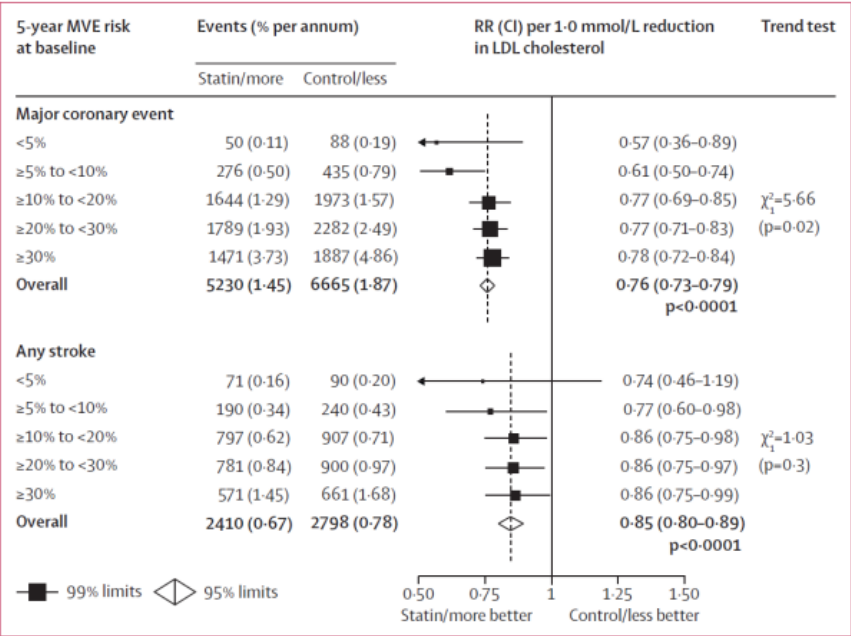


Figure 1: Effects on major coronary events, strokes, coronary revascularisation procedures, and major vascular events per 1.0 mmol/L reduction in LDL cholesterol at different levels of risk
MVE=major vascular event. RR=rate ratio. CI=confidence interval.

Historical prescribing algorithm retired.

Use current risk-specific lipid guidance; see sources below.

Lancet. 2012 Aug 11;380(9841):581-90.
J Am Coll Cardiol. 2014 Jul 1;63:2889-934.

Source-qualified teaching update | 30 September 2026

The 2013 prescribing/risk algorithm is retired from current clinical use. The 2026 guideline uses PREVENT-ASCVD risk assessment for eligible adults aged 30-79 and current risk-specific LDL goals. The retained trial figure is historical evidence, not a replacement prescribing algorithm.

doi.org/10.1161/CIR.0000000000001423
fcm.aulavirtual.unc.edu.ar/pluginfile.php/608531/mod_resource/content/1/Gui%CC%81as%20Dislipemia%202026.pdf

Treatment of Dyslipidemia : Secondary Stroke Prevention

High-Dose Atorvastatin after Stroke or Transient Ischemic Attack

The Stroke Prevention by Aggressive Reduction in Cholesterol Levels (SPARCL) Investigators*

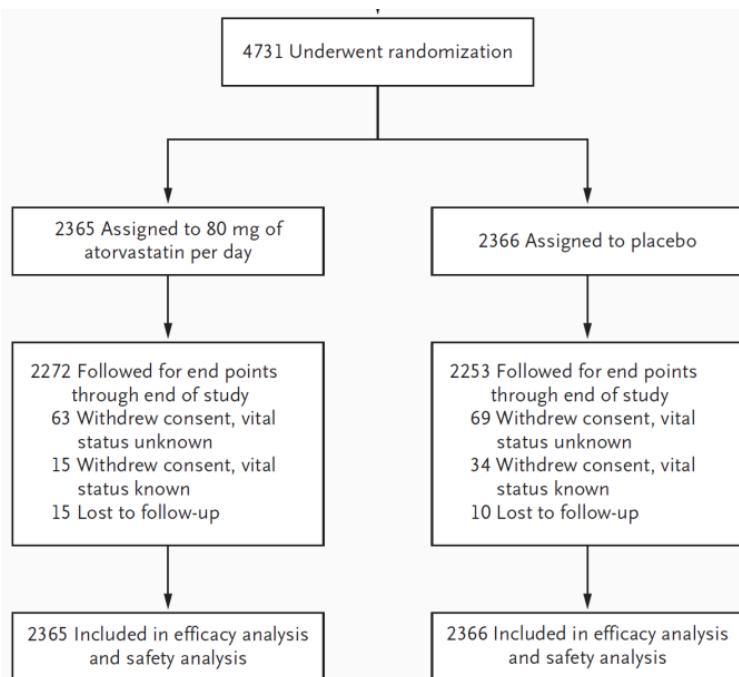


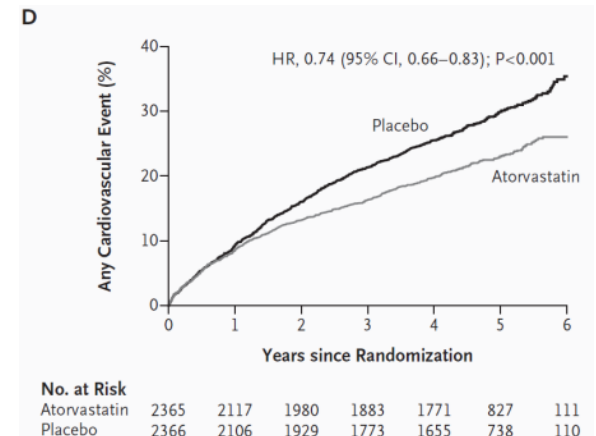
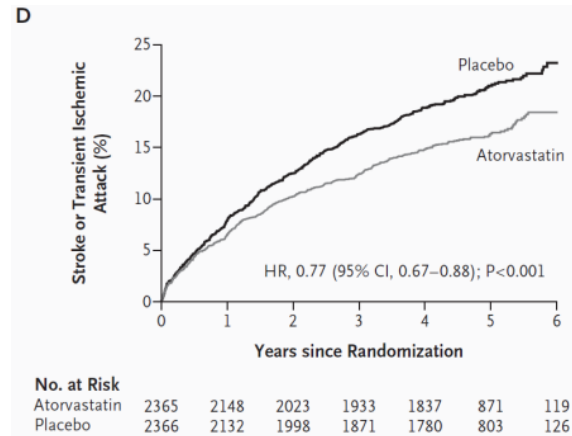
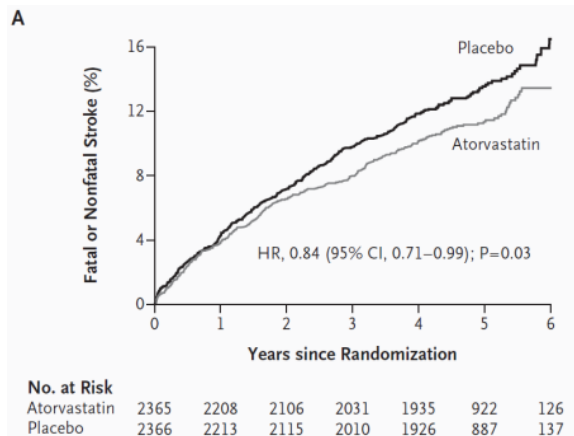
Table 1. Baseline Characteristics of the Patients.*

Characteristic	Atorvastatin (N=2365)	Placebo (N=2366)
Age — yr	63.0±0.2	62.5±0.2
Male sex — no. (%)	1427 (60.3)	1396 (59.0)
Systolic blood pressure — mm Hg	138.9±0.4	138.4±0.4
Diastolic blood pressure — mm Hg	82.0±0.2	81.4±0.2
Body-mass index†	27.5±0.1	27.4±0.1
Entry event — no. (%)		
Stroke	1655 (70.0)	1613 (68.2)
Ischemic	1595 (67.4)	1559 (65.9)
Hemorrhagic	45 (1.9)	48 (2.0)
Other type or not determined	15 (0.6)	6 (0.3)
TIA	708 (29.9)	752 (31.8)
Unknown	2 (0.1)	1 (<0.1)
Time since entry event — days	87.1±1.0	84.3±1.0
Lipids — mg/dL‡		
LDL cholesterol	132.7±0.5	133.7±0.5
HDL cholesterol	50.0±0.3	50.0±0.3
Total cholesterol	211.4±0.6	212.3±0.6
Triglycerides	144.2±1.9	143.2±1.4

Table 2 Selected outcome data from the Stroke Prevention by Aggressive Reduction in Cholesterol Levels (SPARCL) study.⁶

Outcome	Atorvastatin <i>n</i> (%) ^a	Placebo <i>n</i> (%) ^b	Hazard ratio (95% CI)	<i>P</i> value
Primary				
All stroke	265 (11.2)	311 (13.1)	0.84 (0.71–0.99)	0.03
Nonfatal stroke	247 (10.4)	280 (11.8)	0.87 (0.73–1.03)	0.11
Fatal stroke	24 (1.0)	41 (1.7)	0.57 (0.35–0.95)	0.03
Secondary				
Stroke or transient ischemic attack	375 (15.9)	476 (20.1)	0.77 (0.67–0.88)	<0.001
Major coronary event	81 (3.4)	120 (5.1)	0.65 (0.49–0.87)	0.003
Any coronary event	123 (5.2)	204 (8.6)	0.58 (0.46–0.73)	<0.001
Revascularization	94 (4.0)	163 (6.9)	0.55 (0.43–0.72)	<0.001
Any cardiovascular event	530 (22.4)	687 (29.0)	0.74 (0.66–0.83)	<0.001
Death	216 (9.1)	211 (8.9)	1.00 (0.82–1.21)	0.98
Death from cardiovascular disease	78 (3.3)	98 (4.1)	0.78 (0.58–1.06)	0.11

^aTotal number of patients = 2,365. ^bTotal number of patients = 2,366. Abbreviation: *n*, number of patients.



Effects of Statins

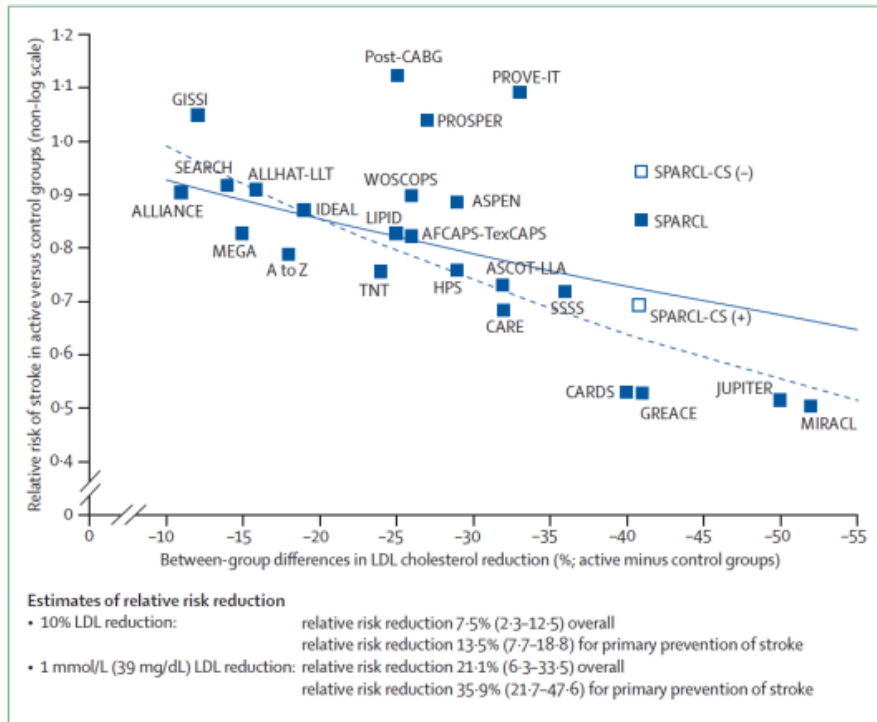


Figure 4: Association between reduction in LDL cholesterol concentration with stroke incidence among the major statins trials

Box 1 Mechanisms of action attributed to statins.

- Upregulation of LDL receptor activity
- Reduction in the entry of LDL cholesterol into the circulation
- Decrease in serum markers of chronic inflammation
- Upregulation of certain genes
- Improvement in endothelial function
- Plaque stabilization
- Antithrombotic effect
- Antioxidant effect
- Anti-inflammatory effect
- Neuroprotection
- Vasoprotection
- Upregulation of endothelial nitric oxide synthase
- Improvement in HDL cholesterol function

Statins and Intracerebral Hemorrhage

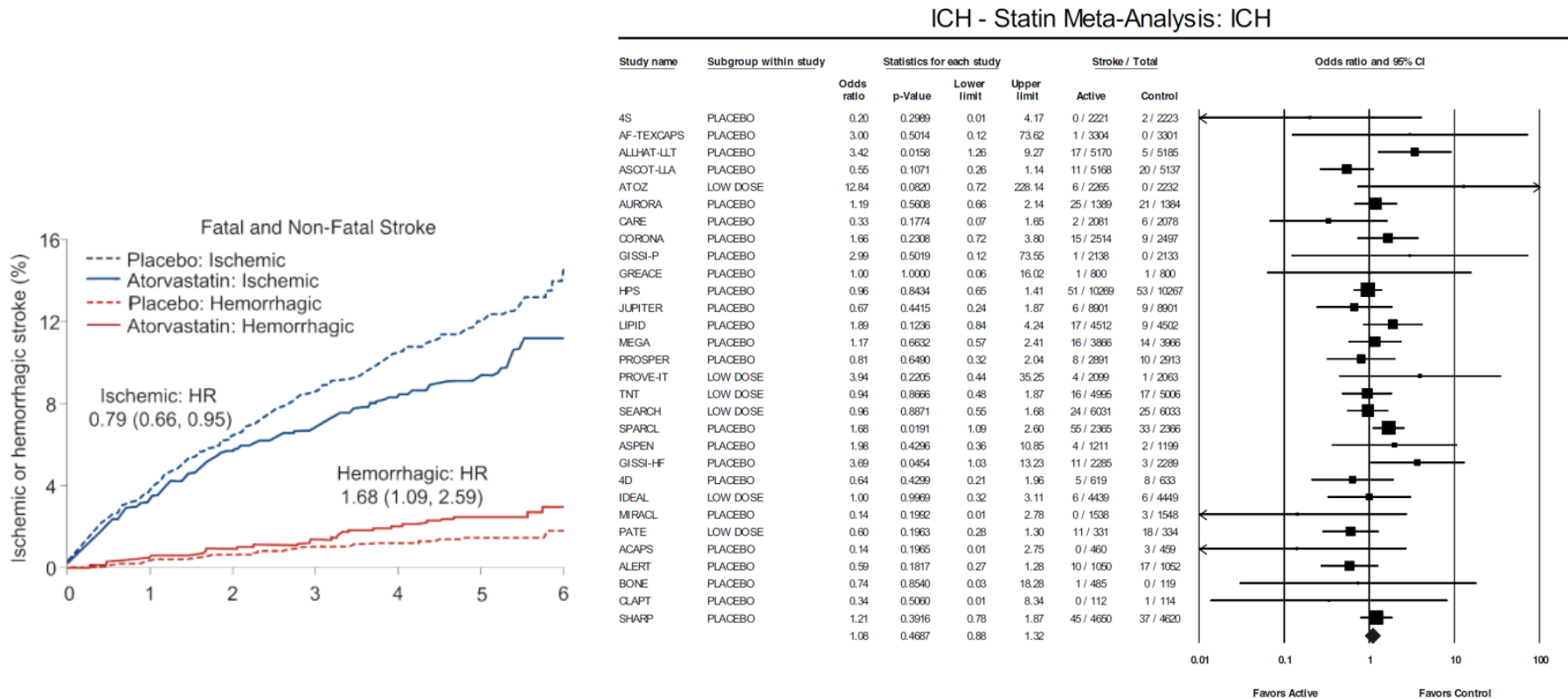


Figure 2. Forrest plot of random-effects meta-analysis of randomized trials of statins and intracerebral hemorrhage.

Statins and Microbleeds: Association

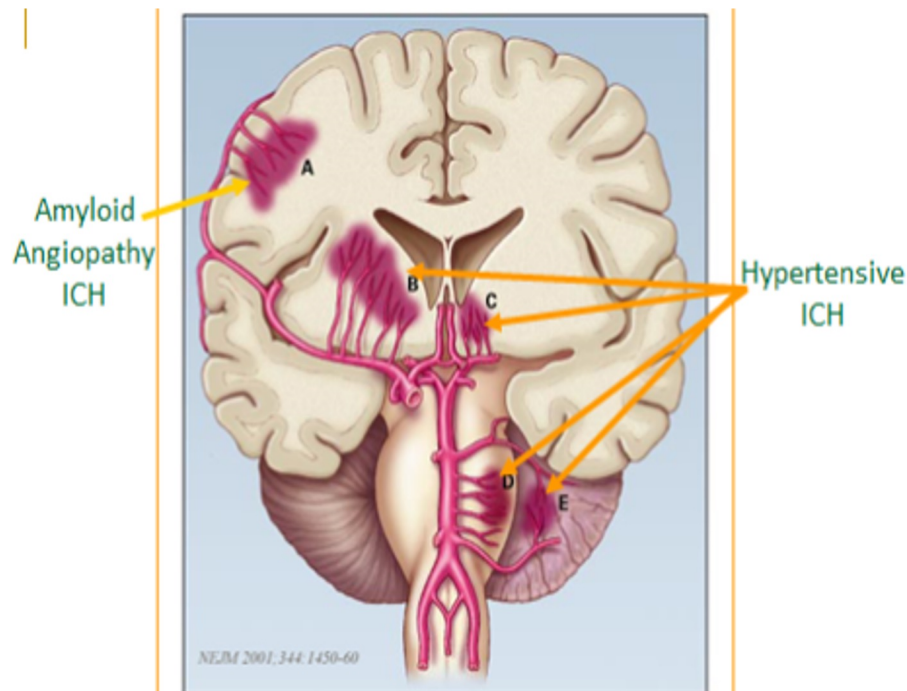
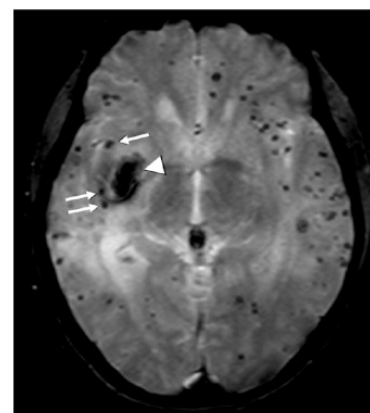


Table 4. Multivariate Logistic Regression Analysis for Presence of Cortico-Subcortical Microbleed

Variable	OR	95% CI	P
Age	1.03	1.00–1.06	0.012*
Male	0.66	0.33–1.31	0.240
Hypertension	0.91	0.42–1.94	0.812
Diabetes	0.42	0.15–1.14	0.090
Dyslipidemia	0.72	0.29–1.75	0.471
Coronary artery disease	0.97	0.35–2.69	0.965
Antiplatelet use	0.75	0.33–1.69	0.496
Statin use	4.15	1.54–11.20	0.005*



Stroke. 2012 Oct;43(10):2677-81.

Source-qualified teaching update | 30 September 2026

This was an observational study of 163 patients with ICH. It associated statin exposure with cortico-subcortical microbleeds (OR 4.15, 95% CI 1.54-11.20); it does not demonstrate recurrent lobar hemorrhage or statin causation.

pubmed.ncbi.nlm.nih.gov/22829545/

Alternative Dyslipidemia Therapies

SUMMARY - ALL TRIALS

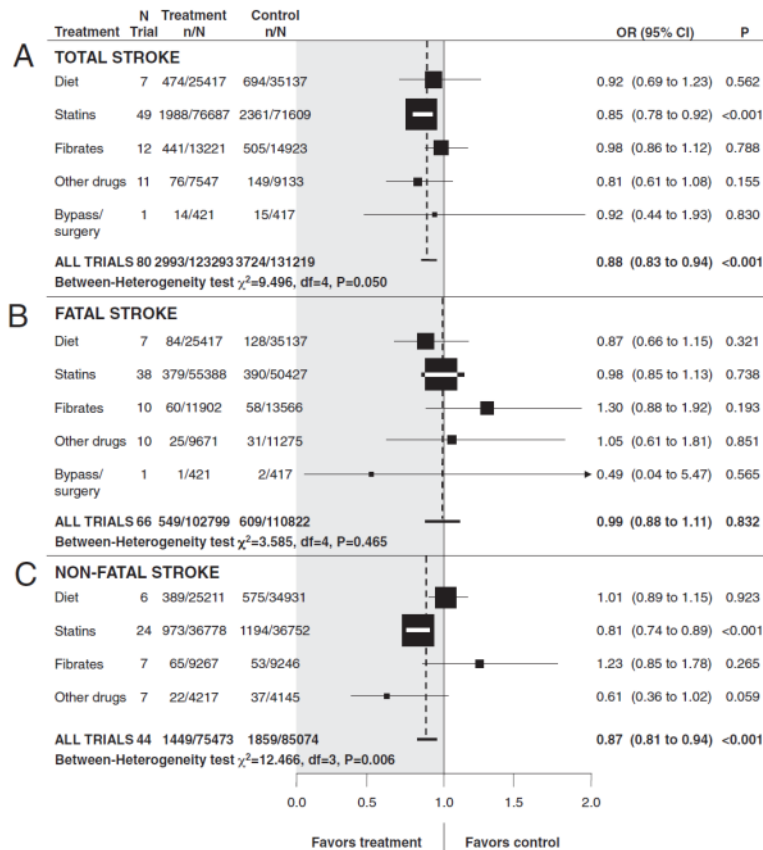


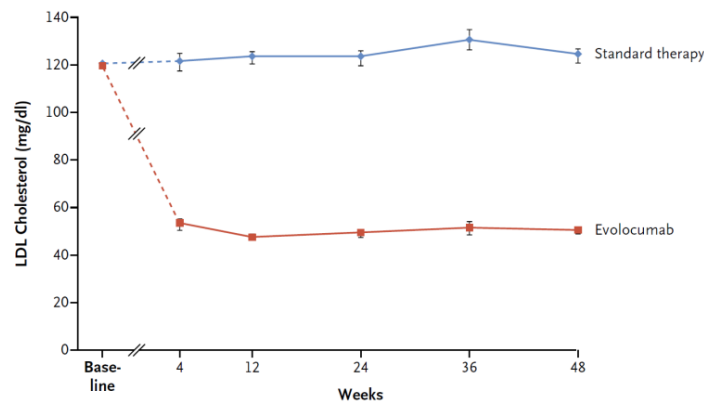
Table 2. Primary, Secondary, and Individual End Points.*

Outcome	Simvastatin Monotherapy (N=9077)	Simvastatin-Ezetimibe (N=9067)	Hazard Ratio (95% CI)	P Value
<i>no. of patients (%)</i>				
Primary end point: death from cardiovascular causes, major coronary event, or nonfatal stroke	2742 (34.7)	2572 (32.7)	0.936 (0.89–0.99)	0.016
Secondary end points				
Death from any cause, major coronary event, or nonfatal stroke	3246 (40.3)	3089 (38.7)	0.95 (0.90–1.0)	0.03
Death from coronary heart disease, nonfatal MI, urgent coronary revascularization ≥ 30 days	1448 (18.9)	1322 (17.5)	0.91 (0.85–0.98)	0.02
Death from cardiovascular causes, nonfatal MI, hospitalization for unstable angina, all revascularization ≥ 30 days, nonfatal stroke	2869 (36.2)	2716 (34.5)	0.95 (0.90–1.0)	0.04
Tertiary end points†				
Any stroke	345 (4.8)	296 (4.2)	0.86 (0.73–1.00)	0.05
Ischemic stroke	297 (4.1)	236 (3.4)	0.79 (0.67–0.94)	0.008
Hemorrhagic stroke	43 (0.6)	59 (0.8)	1.38 (0.93–2.04)	0.11

PCSK9 Inhibitors

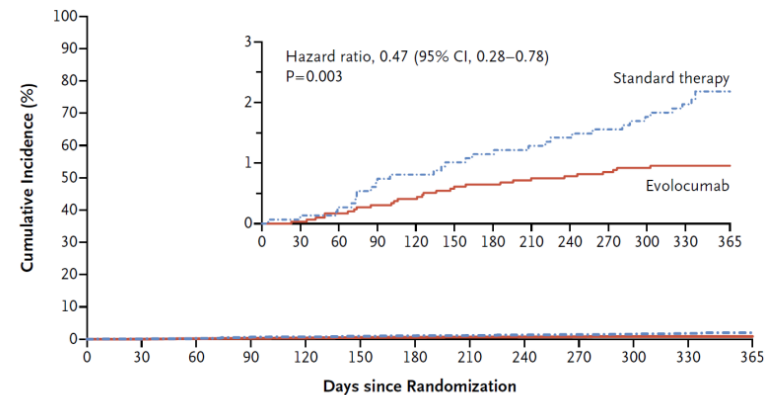
ORIGINAL ARTICLE

Efficacy and Safety of Evolocumab in Reducing Lipids and Cardiovascular Events



No. at Risk						
Standard therapy	1489	394	1388	1376	402	1219
Evolocumab	2976	864	2871	2828	841	2508
Absolute reduction (mg/dl)		60.4	73.4	70.4	72.7	70.5
Percentage reduction		45.3	60.9	58.8	54.0	58.4
P value		<0.001	<0.001	<0.001	<0.001	<0.001

Figure 1. Low-Density Lipoprotein (LDL) Cholesterol Levels.



No. at Risk															
Standard therapy	1489	1486	1481	1473	1467	1463	1458	1454	1447	1438	1428	1361	407		
Evolocumab	2976	2970	2962	2949	2938	2930	2920	2910	2901	2885	2871	2778	843		

Figure 2. Cumulative Incidence of Cardiovascular Events.

N Engl J Med. 2015 Apr 16;372(16):1500-9.

Source-qualified teaching update | 30 September 2026

OSLER was an open-label extension (n=4465, median 11.1 months). Its cardiovascular result was exploratory (HR 0.47, 95% CI 0.28-0.78), alongside LDL lowering. It should not stand alone as current definitive cardiovascular-outcome evidence.

pubmed.ncbi.nlm.nih.gov/25773607/

Guidelines

Historical prescribing algorithms retired.

Current risk-specific lipid guidance and the uncertainty after ICH are summarized below.

Stroke. 2014 Jul;45(7):2160-236.
J Am Coll Cardiol. 2014 Jul 1;63:2889-934.

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The 2013/2014 prescribing algorithms have been retired. Use current risk-stratified lipid guidance. After ICH, AHA/ASA 2022 states that the balance of statin risks and benefits for ICH outcomes/recurrence versus cardiovascular prevention is uncertain (Class IIb, B-NR); this is not an automatic stop or continue rule.

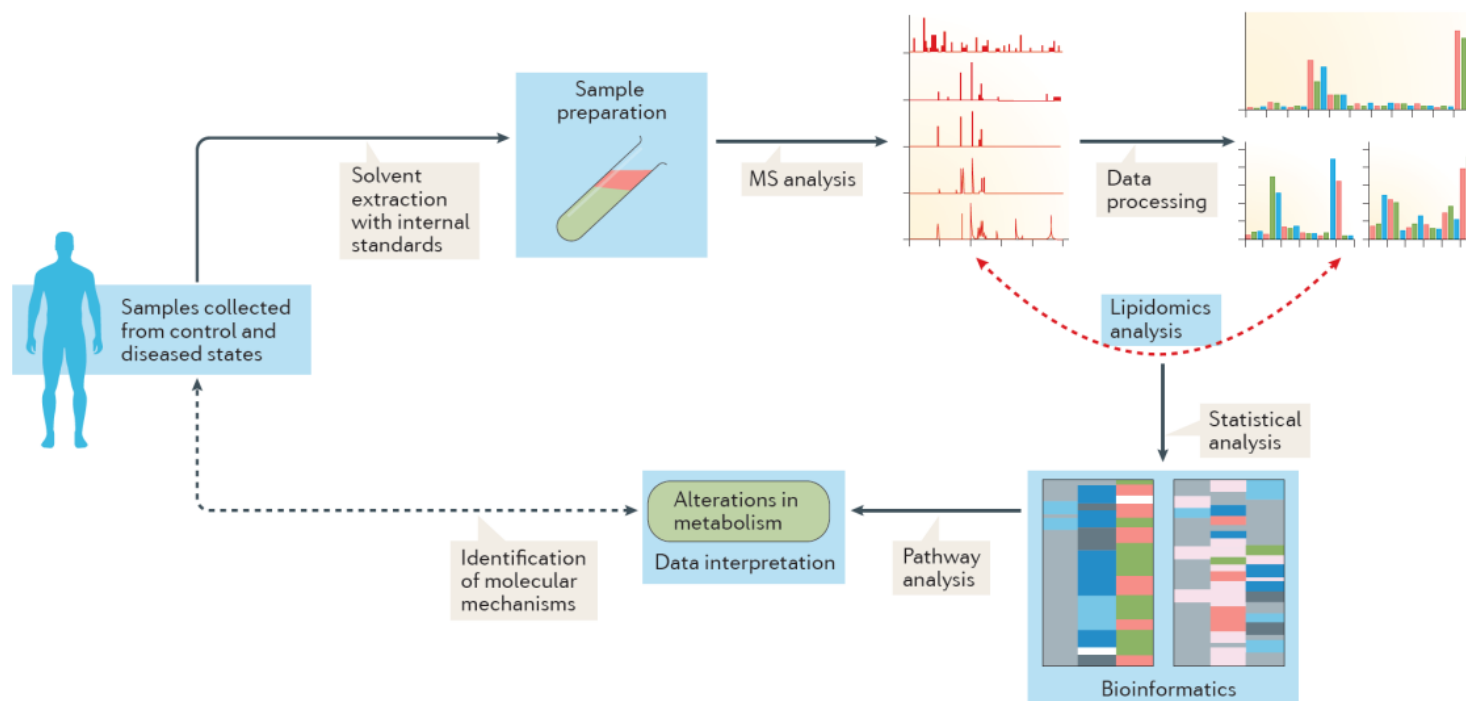
doi.org/10.1161/STR.0000000000000375

doi.org/10.1161/CIR.0000000000001423

doi.org/10.1161/STR.0000000000000407

fcm.aulavirtual.unc.edu.ar/pluginfile.php/608531/mod_resource/content/1/Gui%CC%81as%20Dislipemia%202026.pdf

Lipid Biomarkers: Different Evidence Roles



- Apolipoproteins B & A1
- Lipoprotein (a)
- Lipoprotein-associated phospholipase A2
- HDL sub-fractions

Nat Rev Endocrinol. 2016 Nov;12(11):668-679.

Source-qualified teaching update | 30 September 2026

Lp(a) measurement at least once in adults and selected ApoB assessment are established roles in 2026 guidance. Lp-PLA2 and HDL-subfraction testing remain separate investigational questions.

doi.org/10.1161/CIR.0000000000001423

Apolipoproteins B & A1

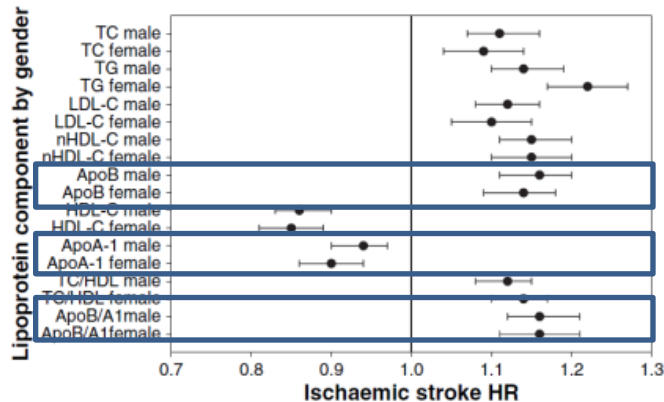


Fig. 1 Ischaemic stroke hazard ratio (HR) for 1 SD difference in lipoprotein components by gender, adjusted for age, acute myocardial infarction, diabetes and hypertension.

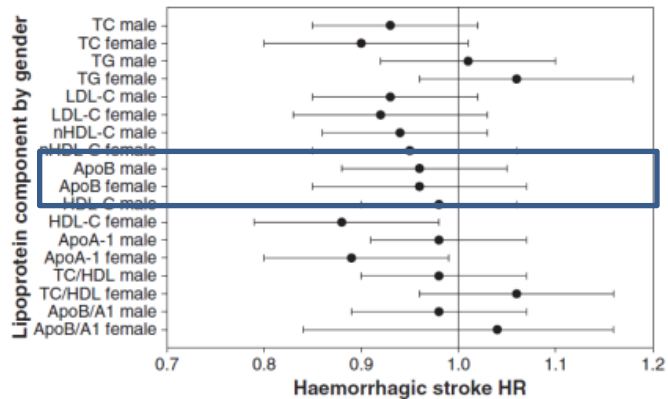
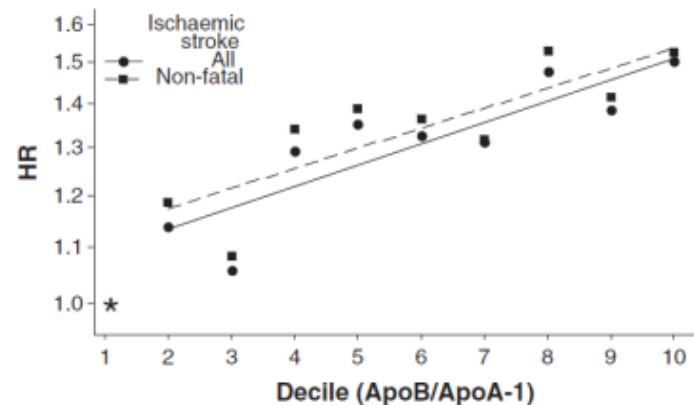


Fig. 2 Haemorrhagic stroke hazard ratio (HR) for 1 SD difference in lipoprotein components by gender, adjusted for age, acute myocardial infarction, diabetes and hypertension.



Means:	0.52	0.63	0.72	0.79	0.89	0.97	1.01	1.10	1.23	1.53
NF #:	189	273	288	375	423	446	451	542	543	606
All #:	210	291	314	402	459	482	498	579	584	651

Fig. 3 Total and nonfatal ischaemic stroke incidence by apoB/apoA-1 in deciles, adjusted for age, gender, total cholesterol, triglycerides, acute myocardial infarction, diabetes and hypertension. Asterisk represents the reference point and is not included in the regression line fits.

Global and regional effects of potentially modifiable risk factors associated with acute stroke in 32 countries (INTERSTROKE): a case-control study

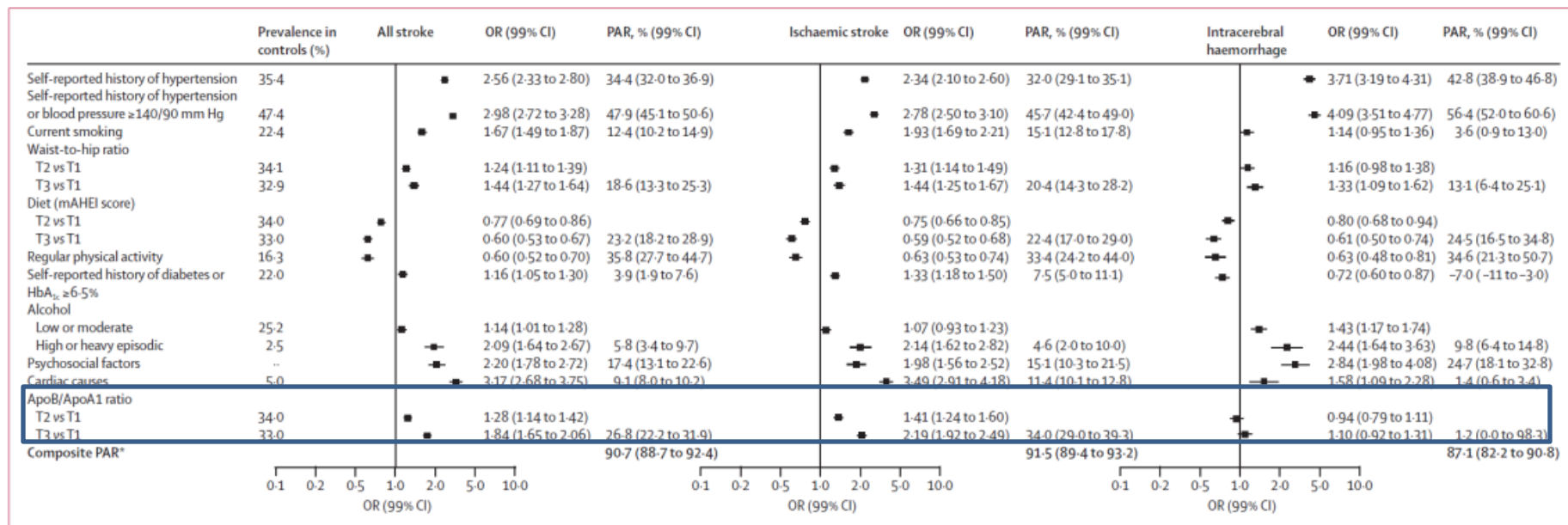


Figure 1: Multivariable analysis of prevalence of risk factors, OR, and PAR for ten risk factors

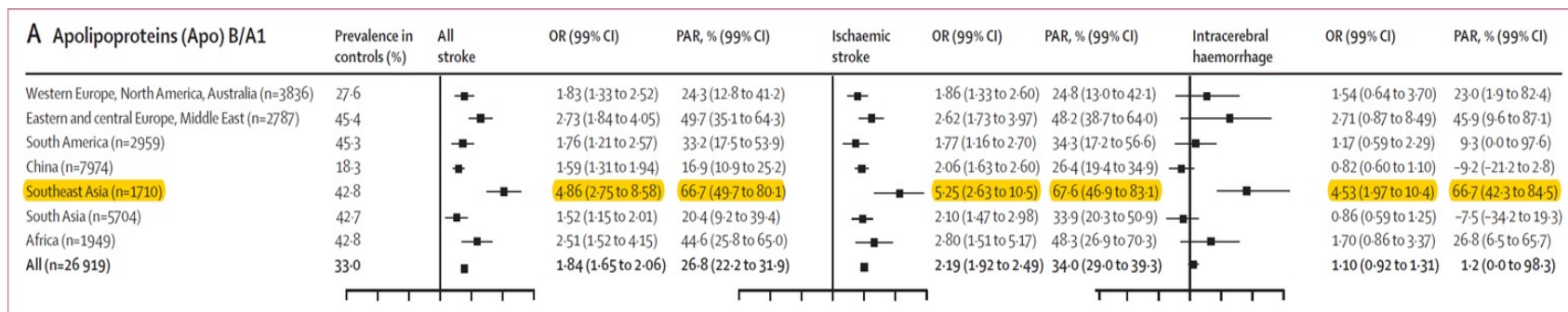


Figure 3: Multivariable analysis by region

Lipoprotein (a)

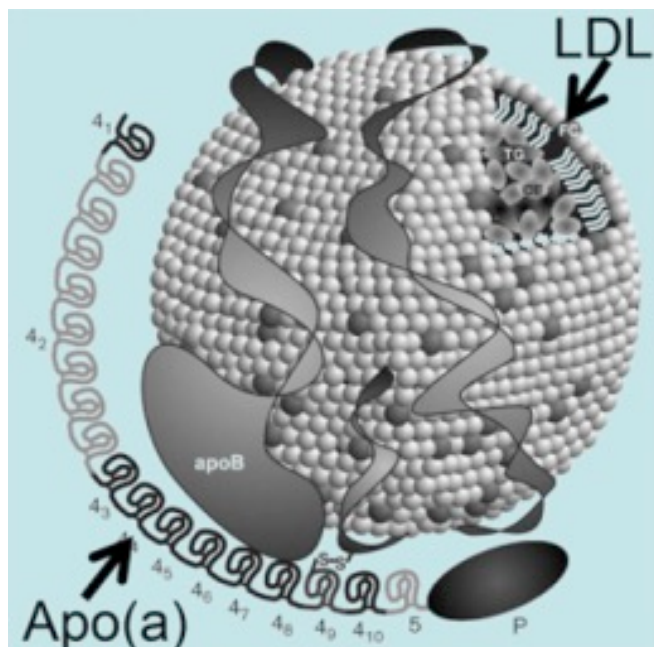


Table 2. ORs for IS and Lp(a) levels

	OR	95% CI	p value
Lp(a) continuous ¹ (per mg/dl)	1.006	1.001–1.01	0.008
Lp(a) continuous ² (per mg/dl)	1.007	1.002–1.01	0.008
Lp(a) ≥30 mg/dl ¹	1.5	1.1–2.0	0.04
Lp(a) ≥30 mg/dl ²	1.6	1.1–2.3	0.03
Lp(a) ≥30 mg/dl ³	1.8	1.2–2.6	0.03

¹ Univariate analysis.

² Adjusted for education level and presence of hypertension, diabetes mellitus, cardiac disease, waist circumference, EtOH, self-report of PAD and smoking, and matched for age, gender and ethnicity.

³ As above but after additional adjustment for fasting lipid and lipoprotein levels.

Table 3. Multivariable models for IS and Lp(a) level ≥30 mg/dl stratified by gender and race/ethnicity

	Mean Lp(a) per mg/dl in cases	OR	95% CI	p value
Gender				
Women	47.4 ± 44	1.4	0.9–2.5	NS
Men	46.6 ± 39	2.3	1.3–4.0	0.01
Race/ethnicity				
Whites	31 ± 24	2.1	0.7–6.6	NS
African Americans	66.9 ± 48	2.7	1.2–6.2	0.04
Hispanics	40.8 ± 36	1.5	0.8–2.5	NS

Stratified models were adjusted for education level and presence of hypertension, diabetes mellitus, cardiac disease, waist circumference, EtOH, self-report of PAD, smoking, fasting lipid and lipoprotein levels. NS = Not significant.

Lipoprotein-Associated Phospholipase A2

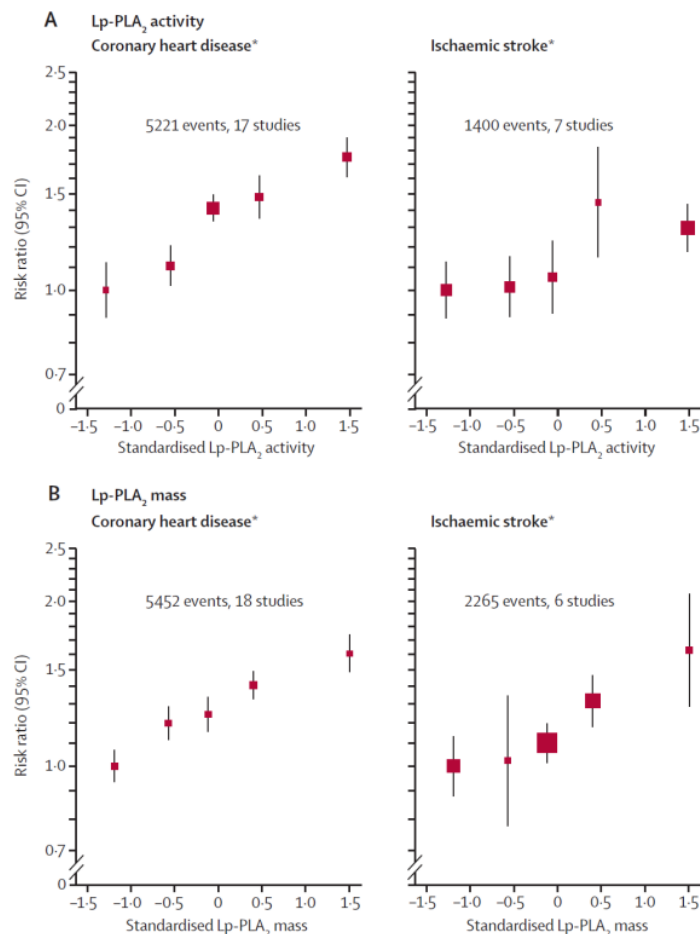


Table 4 Multivariable analysis of factors associated with primary endpoint (ischemic stroke, MI, or death)

Variable	Hazard ratio (95% CI)	p Value
Age per 10 y	1.15 (1.03-1.28)	0.0138
Male	0.88 (0.69-1.12)	0.2963
BMI	1.03 (0.99-1.07)	0.0977
Systolic blood pressure per 20 mm Hg	1.20 (1.09-1.33)	0.0003
History of hypertension	1.23 (0.94-1.59)	0.1256
History of diabetes	1.42 (1.10-1.85)	0.0081
History of MI	0.14 (0.02-1.02)	0.0519
Lp-PLA ₂ -A per 30 nmol/min/mL	1.07 (1.01-1.13)	0.0319

Abbreviations: BMI = body mass index; CI = confidence interval; Lp-PLA₂-A = lipoprotein-associated phospholipase A₂ activity; MI = myocardial infarction.

Lancet. 2010 May 1;375(9725):1536-44.
Neurology. 2015 Nov 3;85(18):1585-91.

Source-qualified teaching update | 30 September 2026

The displayed Lp-PLA₂ activity estimate per 30 nmol/min/mL was HR 1.07 (95% CI 1.01-1.13). After adjustment for LDL, it attenuated to 1.04 (0.97-1.11); this limits an independent-marker interpretation.

pubmed.ncbi.nlm.nih.gov/26311748/

HDL Sub-fractions

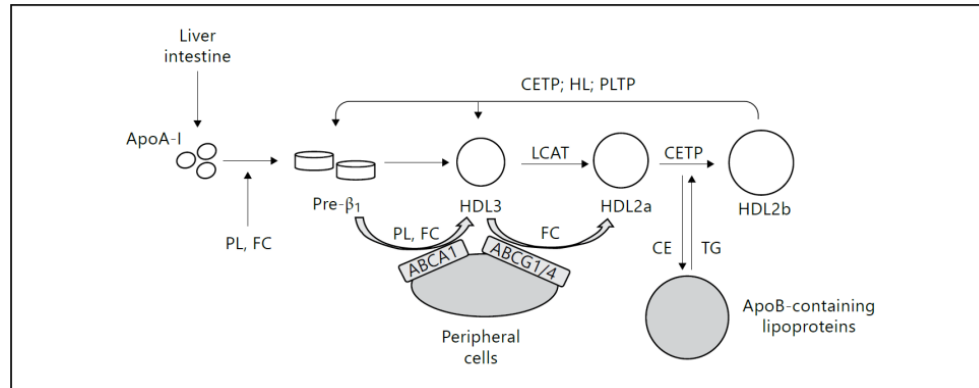


Table 2. Association of HDL2 and HDL3 Subfractions and Total HDL-C With Carotid IMT

IMT Change per 2 SDs	β , SE (n=988)	P Value
HDL2-C		
Model 1	-0.013, 0.005	0.02
Model 2	-0.017, 0.005	0.001
HDL3-C		
Model 1	-0.011, 0.006	0.05
Model 2	0.001, 0.006	0.81
HDL-C		
Model 1	-0.015, 0.006	0.01
Model 2	-0.012, 0.006	0.03

Linear regression model: model 1; adjusted for race/ethnicity, age, sex, low-density lipoprotein-cholesterol, triglycerides, and cholesterol medication use; model 2; adjusted for model 1 and body mass index, smoking, alcohol, physical activity, hypertension, diabetes mellitus, time from baseline to carotid ultrasound. HDL-C indicates high-density lipoprotein-cholesterol; and IMT, intima-media thickness.

Table 3. Association Between the HDL-C Variables and Carotid IMT Stratified by Diabetes Mellitus Status

IMT Change per 2 SDs	β , SE (n=988)	P Value	Interaction P Value
HDL2-C			
Diabetics	-0.043, 0.014	0.003	0.07
Nondiabetics	-0.012, 0.006	0.04	...
HDL3-C			
Diabetics	0.009, 0.014	0.52	1.00
Nondiabetics	0.002, 0.006	0.81	...
HDL-C			
Diabetics	-0.029, 0.012	0.02	0.07
Nondiabetics	-0.008, 0.006	0.24	...

Linear regression model: fully adjusted for race/ethnicity, age, sex, low-density lipoprotein-cholesterol, triglycerides, and cholesterol medication use, body mass index, smoking, alcohol, physical activity, hypertension, and time from baseline to carotid ultrasound. HDL-C indicates high-density lipoprotein-cholesterol; and IMT, intima-media thickness.

Summary

- Positive association between elevated cholesterol and ischemic stroke
 - Large artery atherosclerotic, small vessel subtypes
- Inverse association between cholesterol levels and intracerebral hemorrhage

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Statins reduce recurrent vascular risk in appropriate atherosclerotic/ASCVD populations. This should not be generalized to an identical indication after every cerebral-ischemia mechanism; apply the patient's current risk and indication.

pubmed.ncbi.nlm.nih.gov/16899775/

pubmed.ncbi.nlm.nih.gov/19375663/

doi.org/10.1161/CIR.0000000000001423

Historical research outlook

- Novel lipid-lowering therapies
- Novel lipid biomarkers and cranio-cervical atherosclerosis
- Variation among populations
- Cognitive impairment

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This is a historical research outlook. Current evidence now includes nonstatin outcome therapies and an established Lp(a) screening role; remaining lipidomic, population and cognitive questions should be identified separately as research.

doi.org/10.1161/CIR.0000000000001423

Questions?